

ENGINEERING REBOOT 2026

Day 17 Field Manual

Version 1.0 – Week4 Baseline Publication Edition

MISSION BRIEF

Topic: File Processing & Data Analysis

Objective: Learn how to process records stored in text files and generate useful summaries from stored data.

Expected Outcome: Implement a file-based expense tracker capable of loading expense records, calculating totals, and displaying a summary report.

CONCEPTS

- Processing records
- Line-based processing
- Aggregation
- Running totals
- Sequential file traversal

Example

```
total = 0
```

```
for record in records:
```

```
    total += value
```

EXERCISE 01 – line_counter.py

Objective: Count the number of lines contained within a text file.

Requirements

- Open an existing text file
- Read all records from the file
- Count the number of lines
- Display the total line count
- Program exits cleanly

EXERCISE 02 – word_counter.py

Objective: Count the number of words contained within a text file.

Requirements

- Open an existing text file
- Read file contents
- Count words contained within the file
- Display total word count
- Program exits cleanly

PROJECT / IMPLEMENTATION ASSIGNMENT – expense_tracker.py

Objective: Create a simple expense tracker that processes stored expense records and generates a summary.

Functional Requirements

- Load expense records from a text file
- Process all stored expense entries
- Calculate total expenses
- Display a summary report
- Exit cleanly after processing

Suggested Record Format

25.50

10.00

7.99

15.25

Acceptance Criteria

- Expense records loaded successfully
- All records processed
- Total expenses calculated correctly
- Summary information displayed
- Program exits without errors

VALIDATION MATRIX

Exercise 01

- Existing file loaded successfully
- Line count calculated successfully
- Total line count displayed
- Program exits cleanly

Exercise 02

- Existing file loaded successfully
- Word count calculated successfully
- Total word count displayed
- Program exits cleanly

Project

- Expense records loaded
- All records processed
- Totals calculated correctly
- Summary displayed
- Program exits cleanly

DELIVERABLES

Exercises

- line_counter.py
- word_counter.py

Project

- expense_tracker.py

Documentation

- Notes.md
- Journal.md

EVIDENCE CHECKLIST

- Code Evidence
- Output Evidence
- Notes Updated
- Journal Updated

SOURCE CONTROL

- Commit completed
- Push completed
- Remote sync verified

PREVIEW OF NEXT TOPIC

Day 18 – CSV Fundamentals

- CSV structure
- CSV reading
- CSV writing
- Project: student_grades_csv.py

DAILY SUCCESS CRITERIA

- line_counter.py implemented and validated
- word_counter.py implemented and validated
- expense_tracker.py implemented
- Totals calculated correctly
- Summary displayed correctly
- Notes.md updated
- Journal.md updated
- Code evidence captured
- Output evidence captured
- Commit completed
- Push completed
- Remote synchronization verified

IMPLEMENTATION STRATEGY

The focus of Day 17 is processing data that already exists in storage. Emphasis should be placed on record loading, iteration, aggregation, counting, and summary generation. Implementations should remain aligned with stated requirements and acceptance criteria.

EXECUTION FOCUS

- Correct file reading
- Proper record iteration
- Accurate counting operations
- Accurate total calculations
- Clean program execution

READINESS CHECK

- Text file reading understood from Day 16
- Basic loops understood
- Variables and accumulation understood
- File paths understood
- Development environment operational